
# Operational Modal Analysis of tie-rods

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# Operational Modal Analysis

OMA is a technique used to determine the modal parameters of a structure or a mechanical system under its actual operating conditions.

Unlike traditional modal analysis, OMA does not require artificial excitation but relies on the natural excitation technique (NeXT), exploiting ambient vibrations or operational forces present in the environment.

In particular, the results are as good as the following hypothesis are satisfied:

- Input forces must be random and uncorrelated
- Forcing inputs are assumed to be uniformly distributed in space.

OMA can be carried out both in frequency domain and in time domain.

In the following slides the FDD (Frequency Domain Decomposition) algorithm is presented.

# Why OMA is important in health-monitoring of tie-rods

Structural health monitoring systems are used to determine if an operating structure still meets performance and safety standards despite aging and degradation.

A reduction in the performances can be caused both by damage or by environmental variations. The main difficulty consists in distinguishing between the two effects.

In particular, this can be done by examining changes in the modal parameters of multiple vibration modes together.

In order to estimate modal parameters, OMA can be used.

*Figure: examples of tie-rods applications*

# Objectives of the activity

The aims of the activity are reported below:

1. Carry out the Operational Modal Analysis for all the three conditions (healthy, damage 1 and damage 2), i.e. find the eigenfrequencies, the mode shapes and the damping ratios

2. Compute MAC values in order to compare the mode shapes obtained in different conditions

# Setup

The system consists in a clamped-clamped tie-rod, equipped with 4 accelerometers placed at different positions along the structure:

- $\xi_1$ = 5%
- $\xi_2$ = 33,3 %
- $\xi_3$ = 50 %
- $\xi_4$ = 90%

The signal acquisition has been carried out in 3 different situations:

- healthy condition ($f_s = 512\ Hz$)
- damage 1 condition ($f_s = 1024\ Hz$)
- damage 2 condition ($f_s = 1024\ Hz$)

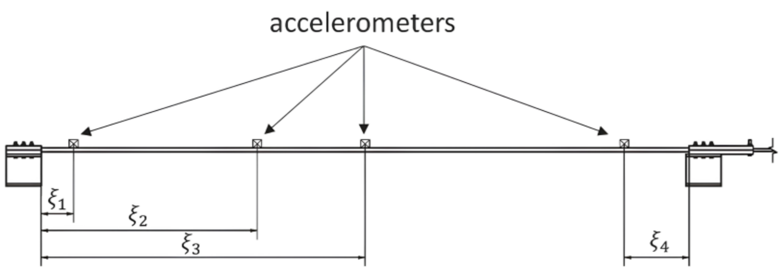



*Figure: experimental setup*

# Setup

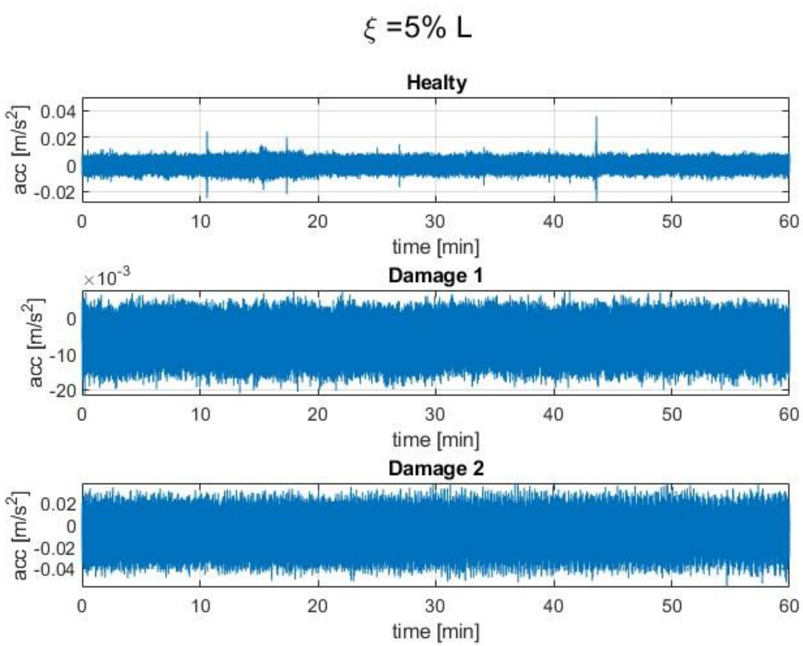



*Figure: signals acquired by the sensors placed at ξ = 0.05 L*

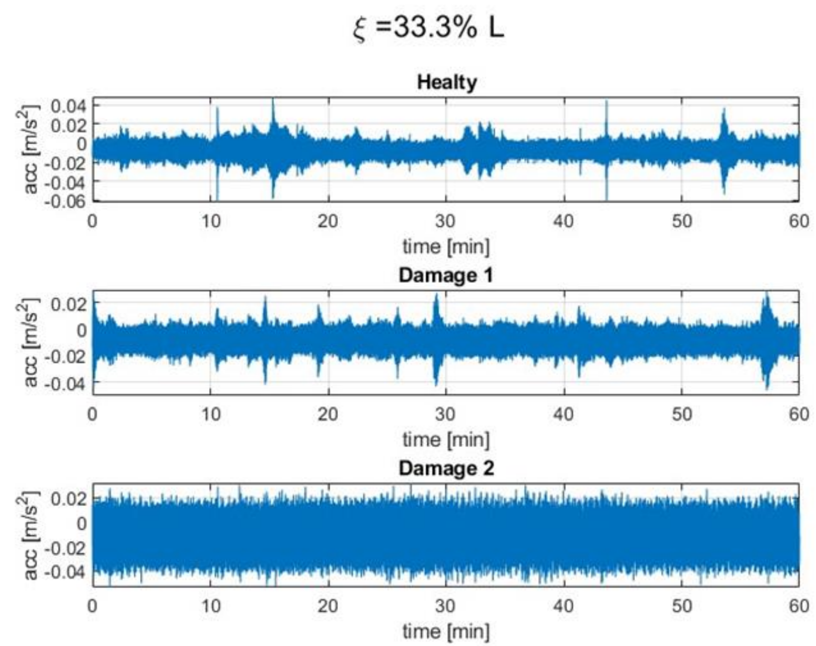



*Figure: signals acquired by the sensors placed at ξ = 0.33 L*

# Setup

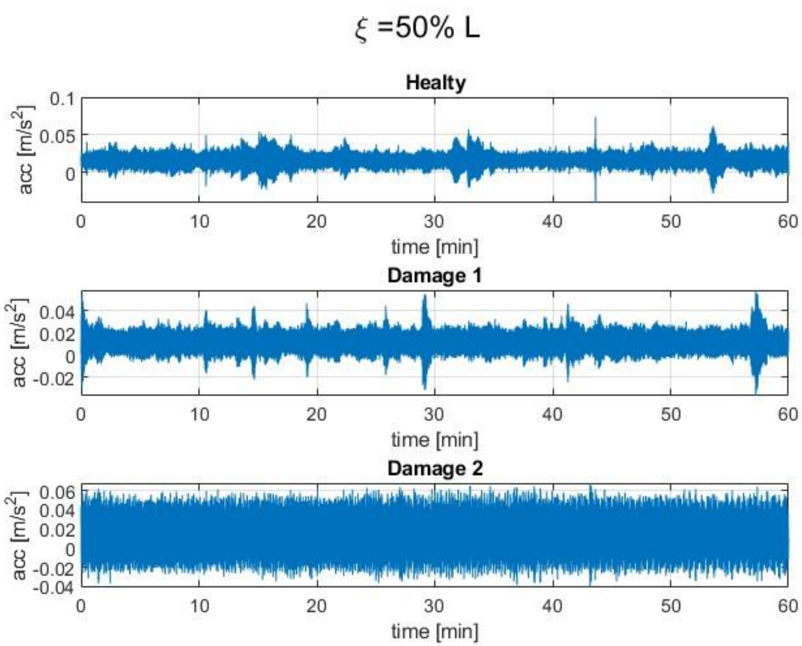



*Figure: signals acquired by the sensors placed at $\xi = 0.5\ L$*

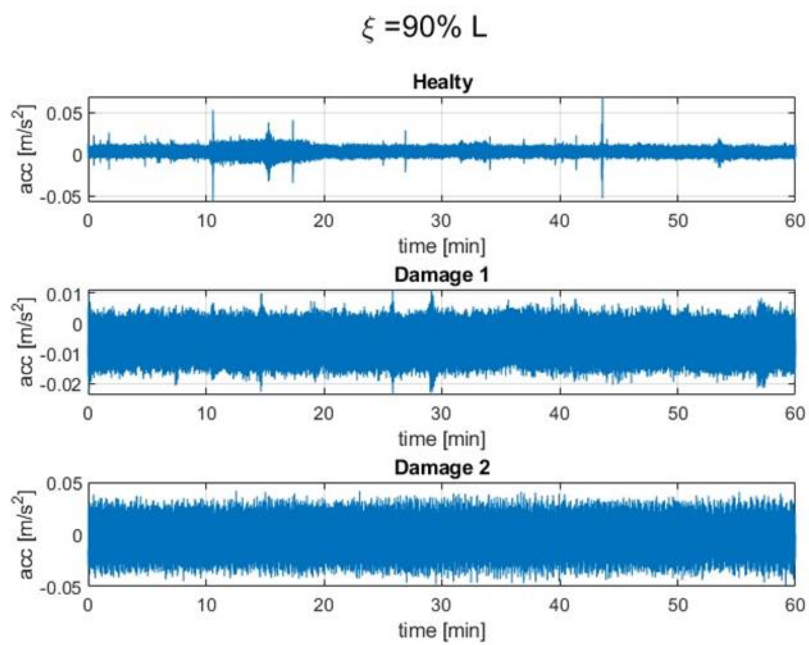



*Figure: signals acquired by the sensor placed at $\xi = 0.9\ L$*

# Stationarity

The power spectral density and the cross spectral density are computed through a proper average process carried out in frequency domain.

This procedure is obtained averaging the cross spectra estimates calculated from each time history obtained subdividing each signal into sub-records, computed using an overlap of 66%.

It is possible to do so since all the measured signals are random and stationary.

Stationarity of the signals has been checked computing the RMS over the different sub-records.

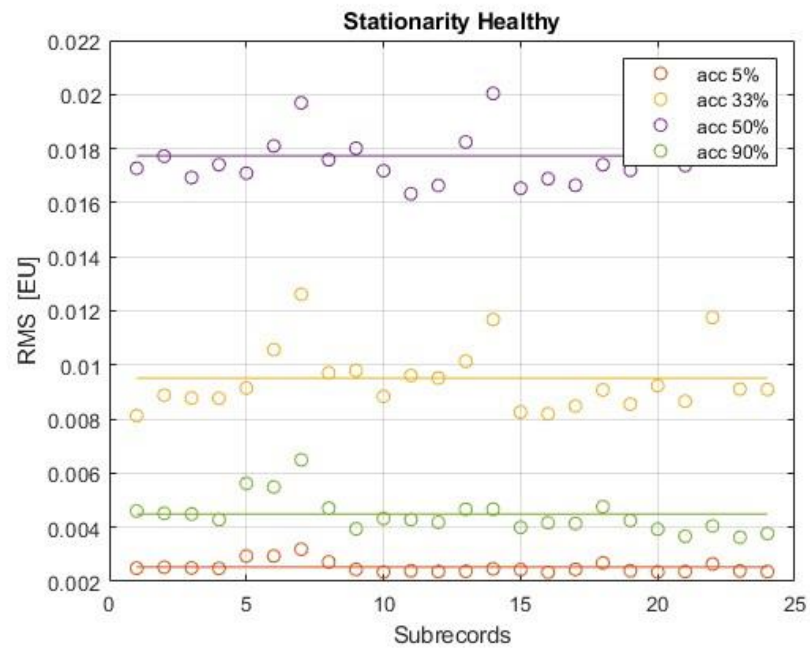



*Figure: stationarity check in healthy condition*

# Stationarity

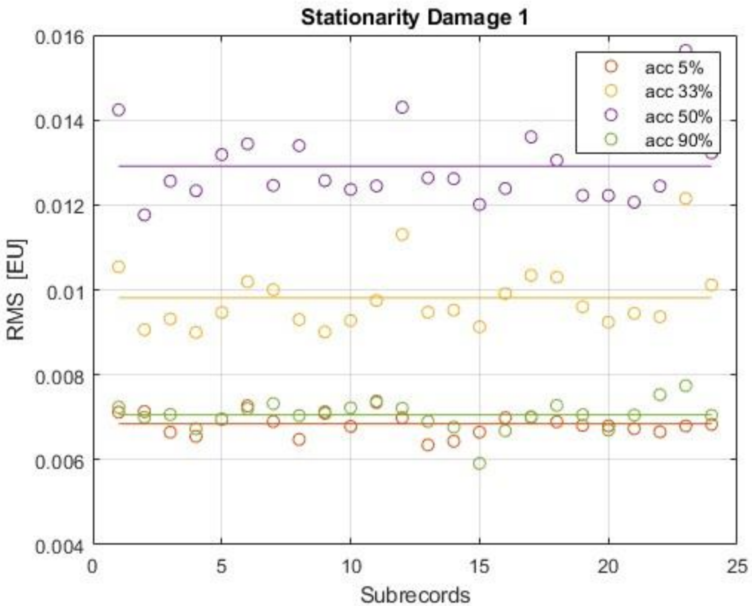



*Figure: stationarity check in damage 1 condition*

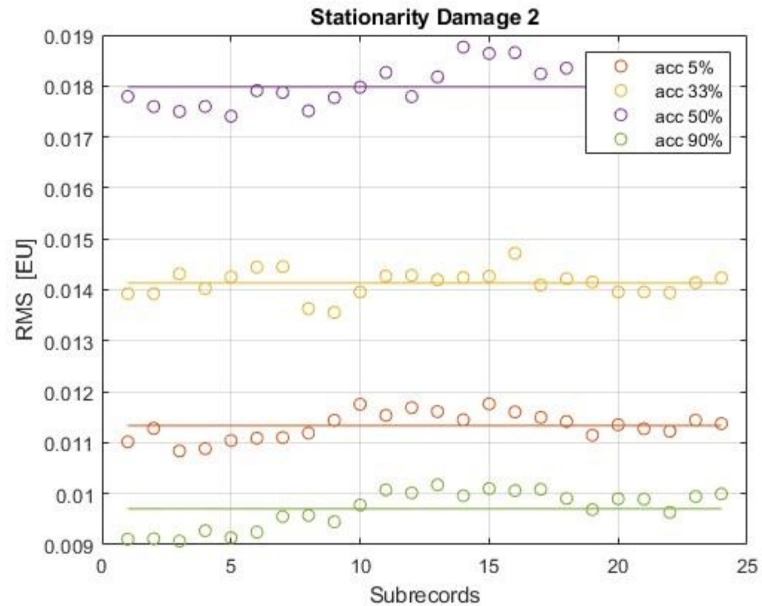



*Figure: stationarity check in damage 2 condition*

# 1. Cross Spectral matrix

The first step in the FDD algorithm consists in expressing the cross spectral density matrix in terms of modal parameters.

According with the modal approach, the system response can be express as:

$$\{y(t)\} = [\phi]\{q(t)\}$$

Moving to frequency domain applying the DFT, we obtain:

$$[Y(f)] = \mathcal{F}\{y(t)\} = [\phi]\{Q(f)\}$$

The cross spectral density matrix can be obtained as:

$$\left[G_{yy}(f)\right] = \{Y(f)\}\{Y(f)\}^H$$

Switching to modal coordinates we obtain:

$$\left[G_{yy}(f)\right] = \{Y(f)\}\{Y(f)\}^H = [\phi]\{Q(f)\}\{Q(f)\}^H[\phi]^H = [\phi]\left[G_{qq}(f)\right][\phi]^H$$

# 2. Singular Value Decomposition

Decomposing $\left[G_{yy}(f)\right]$ using Singular Value Decomposition we obtain:

$$\left[G_{yy}(f)\right] = [U][S][V]$$

where:

- $[U]_{4x4} = [\phi] = [\{\varphi_1\} \ \ \{\varphi_2\} \ \ \ldots \ \ \{\varphi_n\}]$

- $[S]_{4x4} = \left[G_{qq}(f)\right] = \begin{bmatrix} G_{q_1q_1}(f) & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & G_{q_nq_n}(f) \end{bmatrix}$

- $[V]_{4x4} = [\phi]^H = [\phi]^T = \begin{bmatrix} \{\varphi_1\}^T \\ \{\varphi_2\}^T \\ \vdots \\ \{\varphi_n\}^T \end{bmatrix}$ (assuming real mode shapes)

# 3. Singular value curves

- Singular value decomposition must be carried out frequency by frequency
- S.v. curves show peaks around resonances (approximation to SDOF systems)
- Results are shown for different time window lengths.

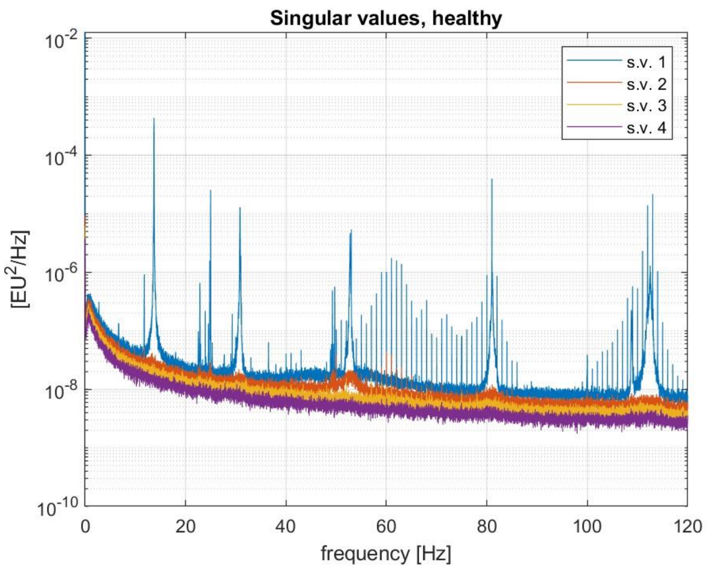



*Figure: singular value curves,* $T = 150\ s$

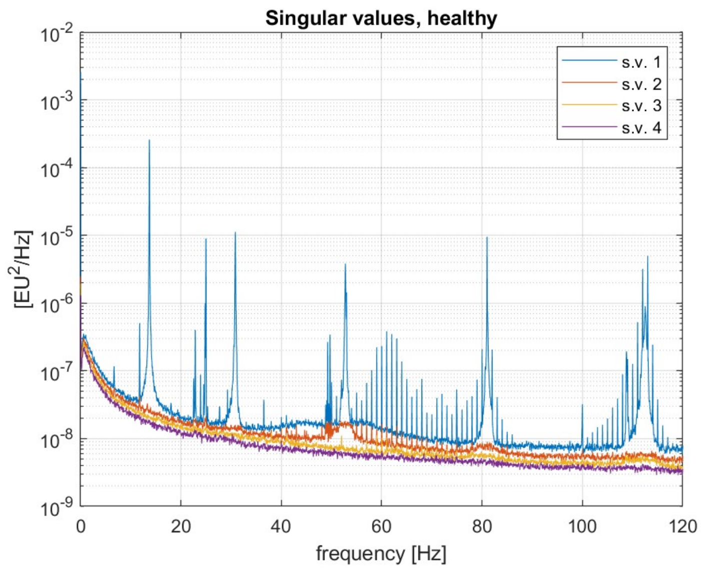



*Figure: singular value curves,* $T = 30\ s$

# 3. Singular value curves

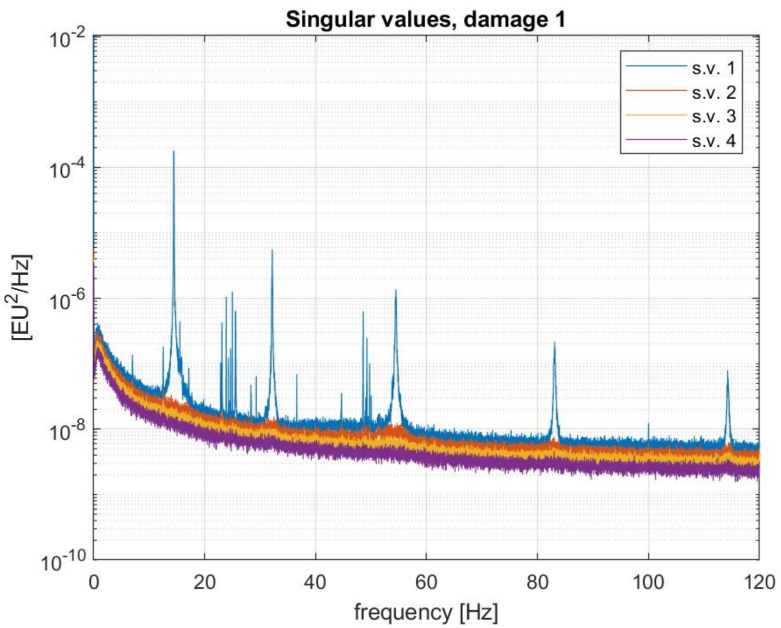



*Figure: singular value curves,* $T = 150\ s$

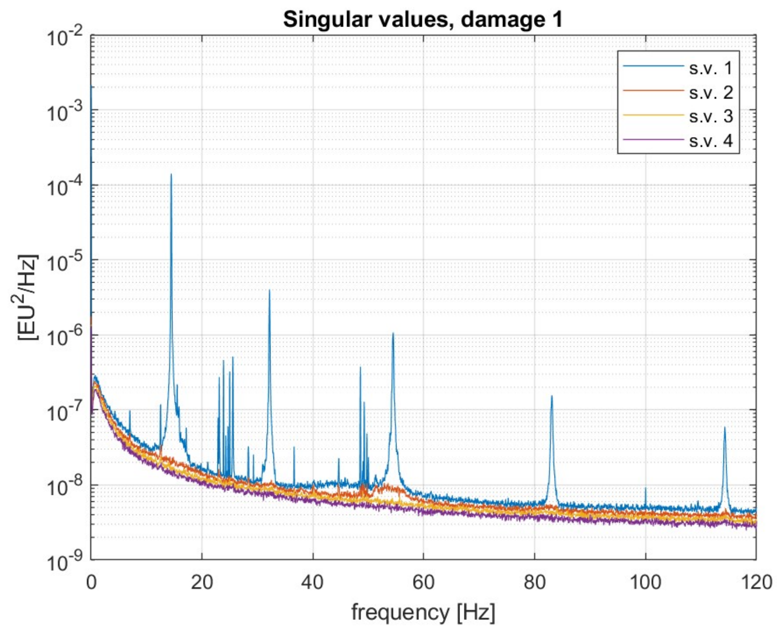
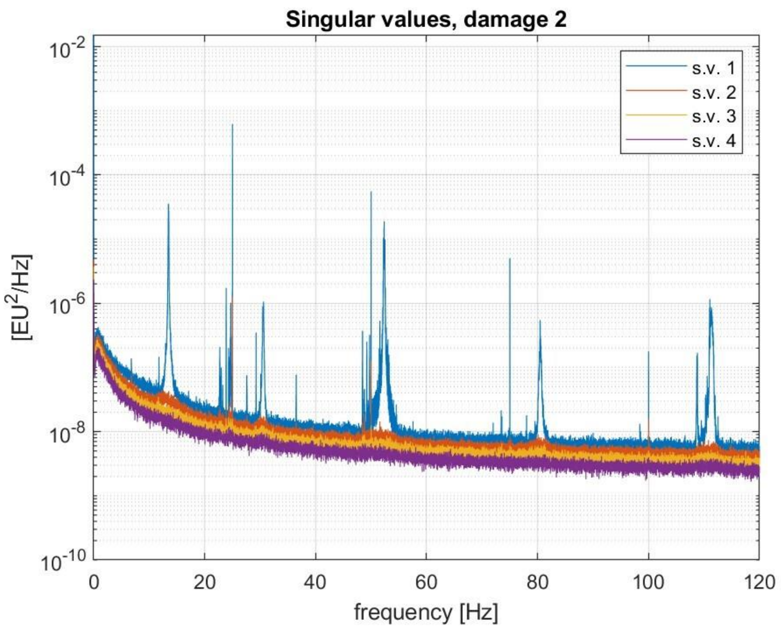



*Figure: singular value curves,* $T = 30\ s$

# 3. Singular value curves

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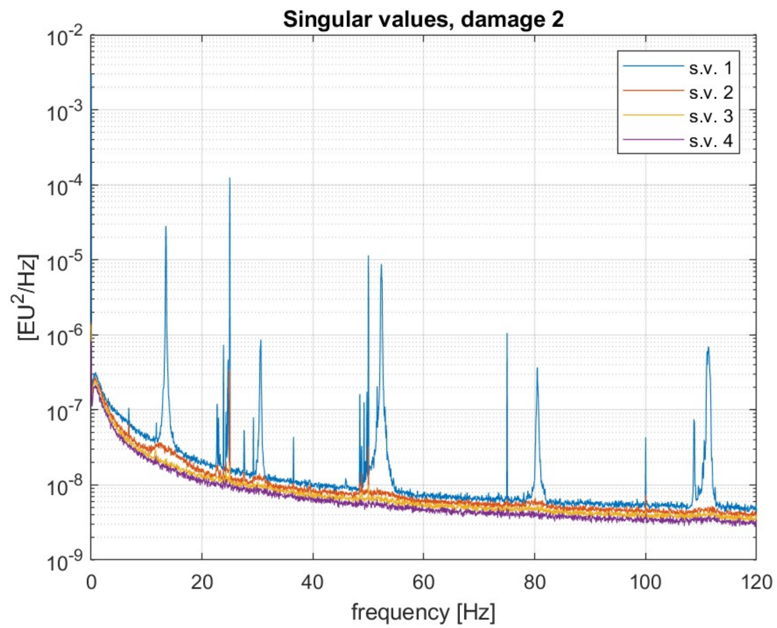



*Figure: singular value curves,* $T = 30\ s$

# 3. Singular value curves

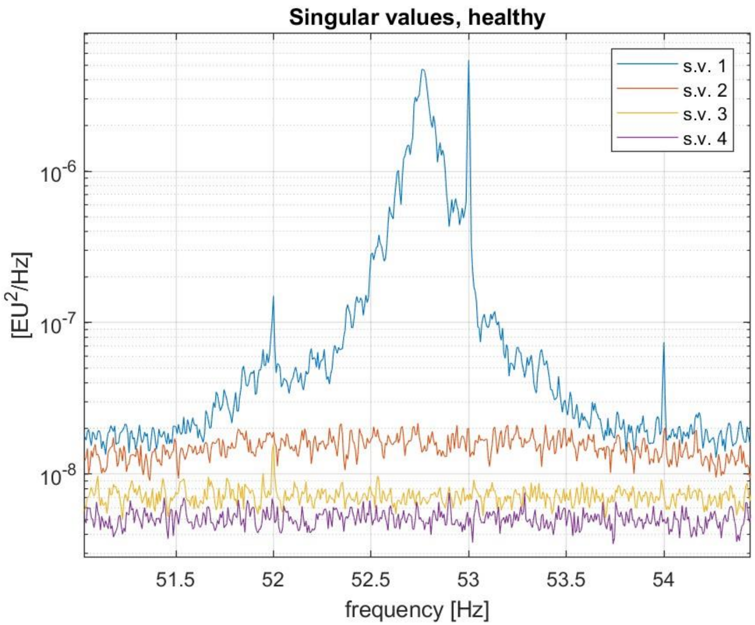



*Figure: third resonance peak in healthy condition, T = 150 s*

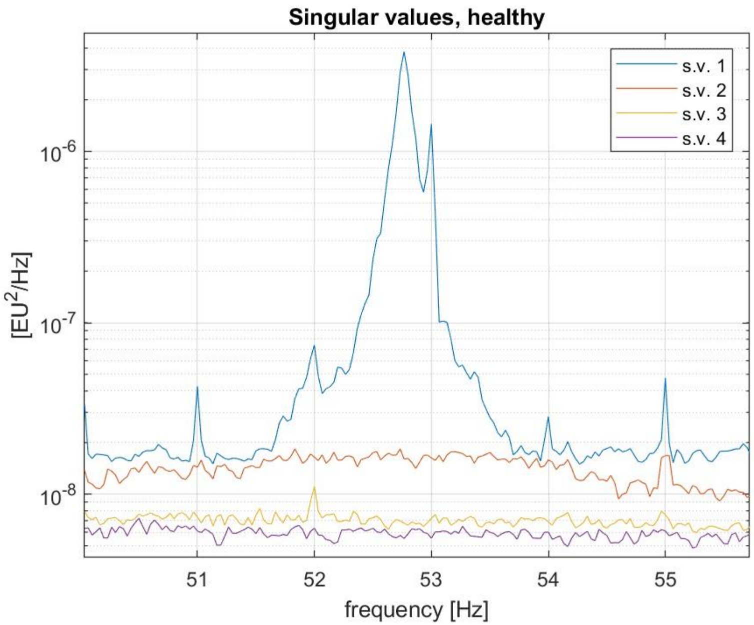



*Figure: third resonance peak in healthy condition, T = 30 s*

# 3. Singular value curves

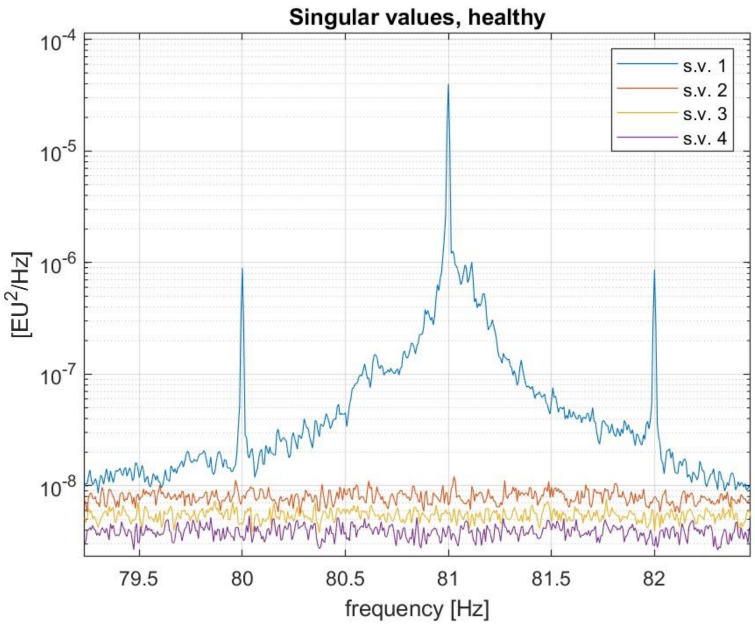



*Figure: fourth resonance peak in healthy condition, T = 150 s*

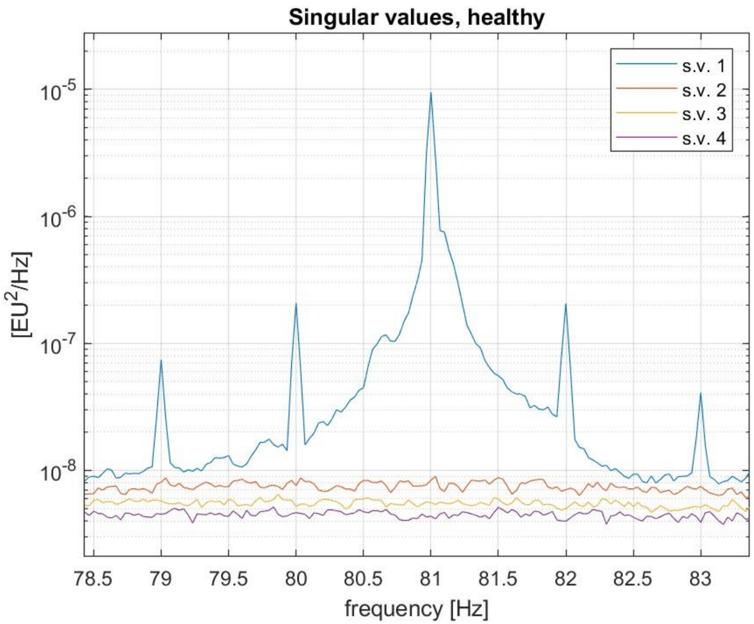



*Figure: fourth resonance peak in healthy condition, T = 30 s*

# 4. Eigenfrequencies identificaton

The eigenfrequencies have been obtained looking at the peaks of the singular value curves, resulting in the following values:

| | Healthy | Damage 1 | Damage 2 |
|---|---|---|---|
| $\omega_1$ | 13.71 $Hz$ | 14.41 $Hz$ | 13.46 $Hz$ |
| $\omega_2$ | 30.84 $Hz$ | 32.17 $Hz$ | 30.60 $Hz$ |
| $\omega_3$ | 52.76 $Hz$ | 54.47 $Hz$ | 52.34 $Hz$ |
| $\omega_4$ | 81.08 $Hz$ | 83.07 $Hz$ | 80.44 $Hz$ |

Once the natural frequencies have been identified, it is possible to obtain the mode shape $\{\varphi_r\}$ by simply extracting the first column of $[\phi]$ when $\omega = \omega_r$

# 5. Mode shapes

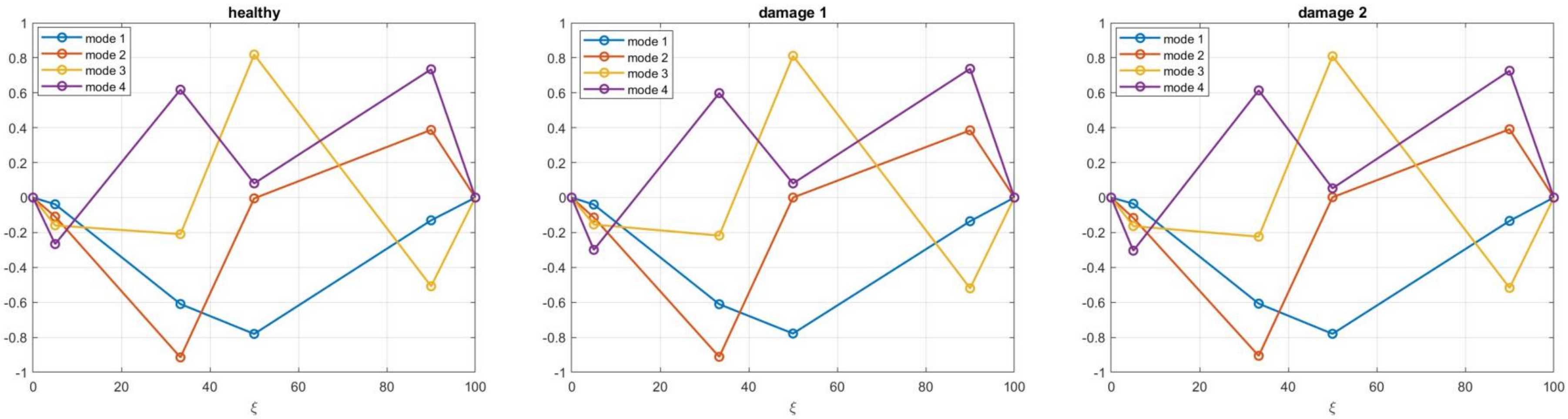



*Figure: Mode shapes computed in healthy condition (left), in damage 1 condition (center), and in damage 2 condition (right)*

# 5. Mode shapes

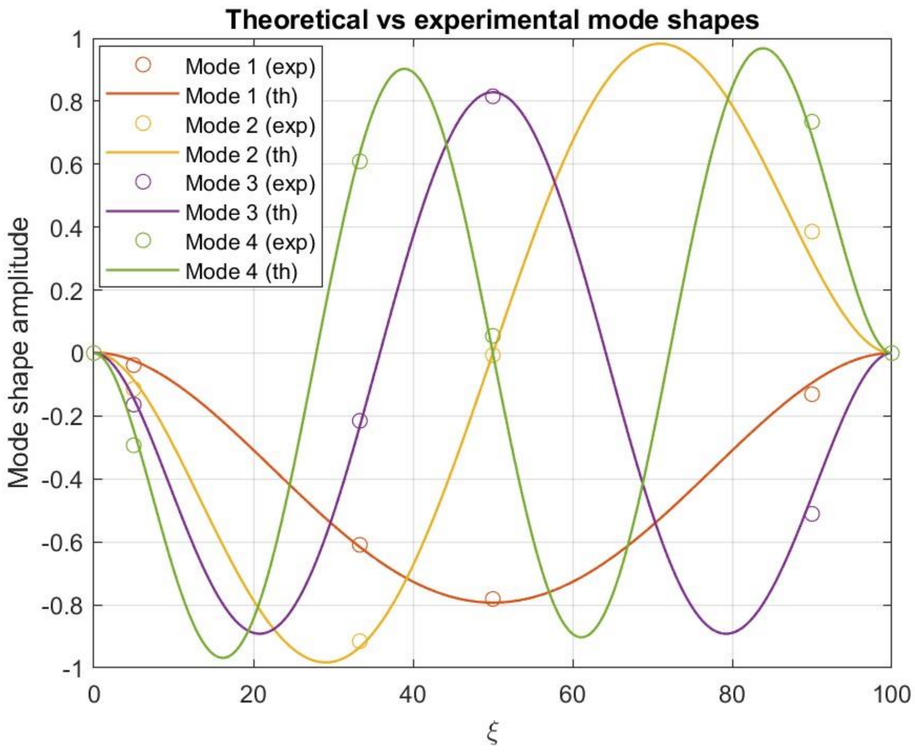



*Figure: comparison between theoretical and experimental mode shapes*

# MAC computation

Once mode shapes have been extracted, it is possible to compute the MAC, comparing the modes obtained in all the 3 conditions with the ones resulting from the healthy condition

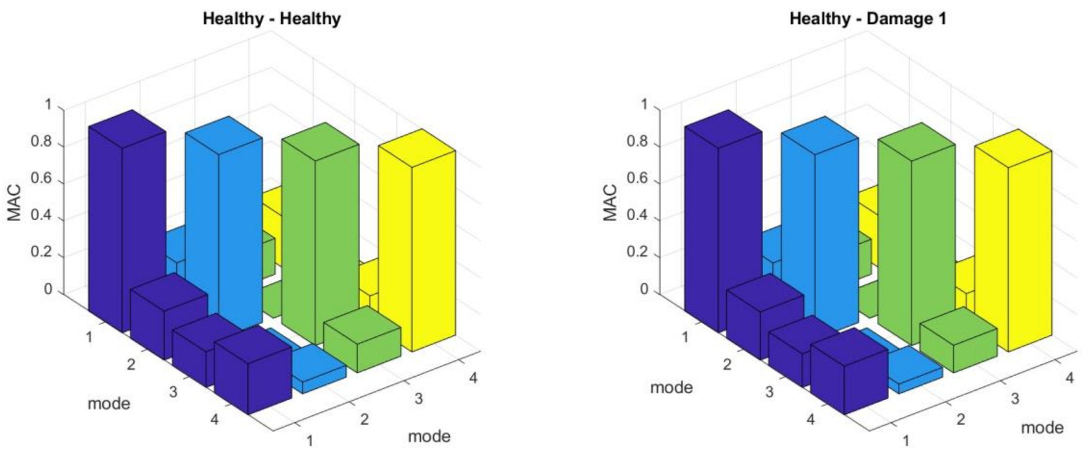



*Figure: MAC values*

# FRFs reconstruction

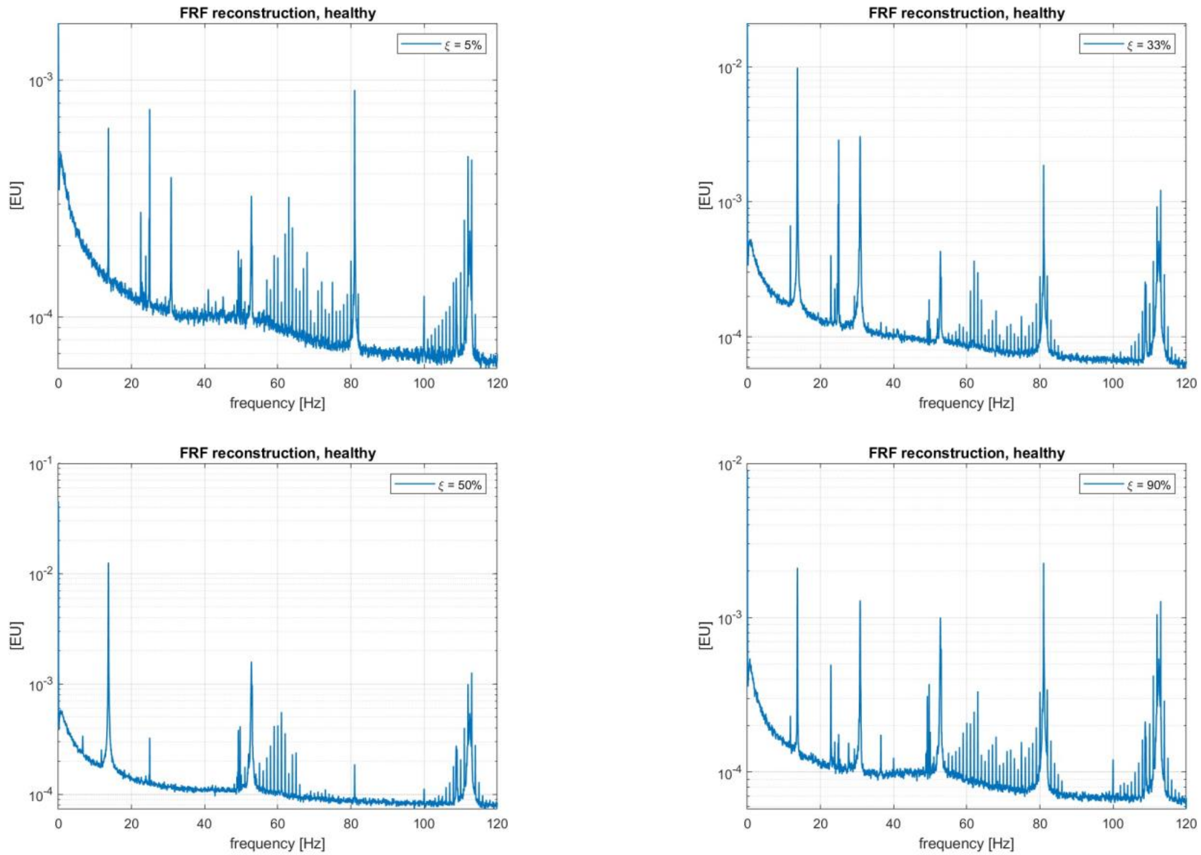



*Figure: reconstruction of the FRFs related to each sensor, healthy*

# FRFs reconstruction

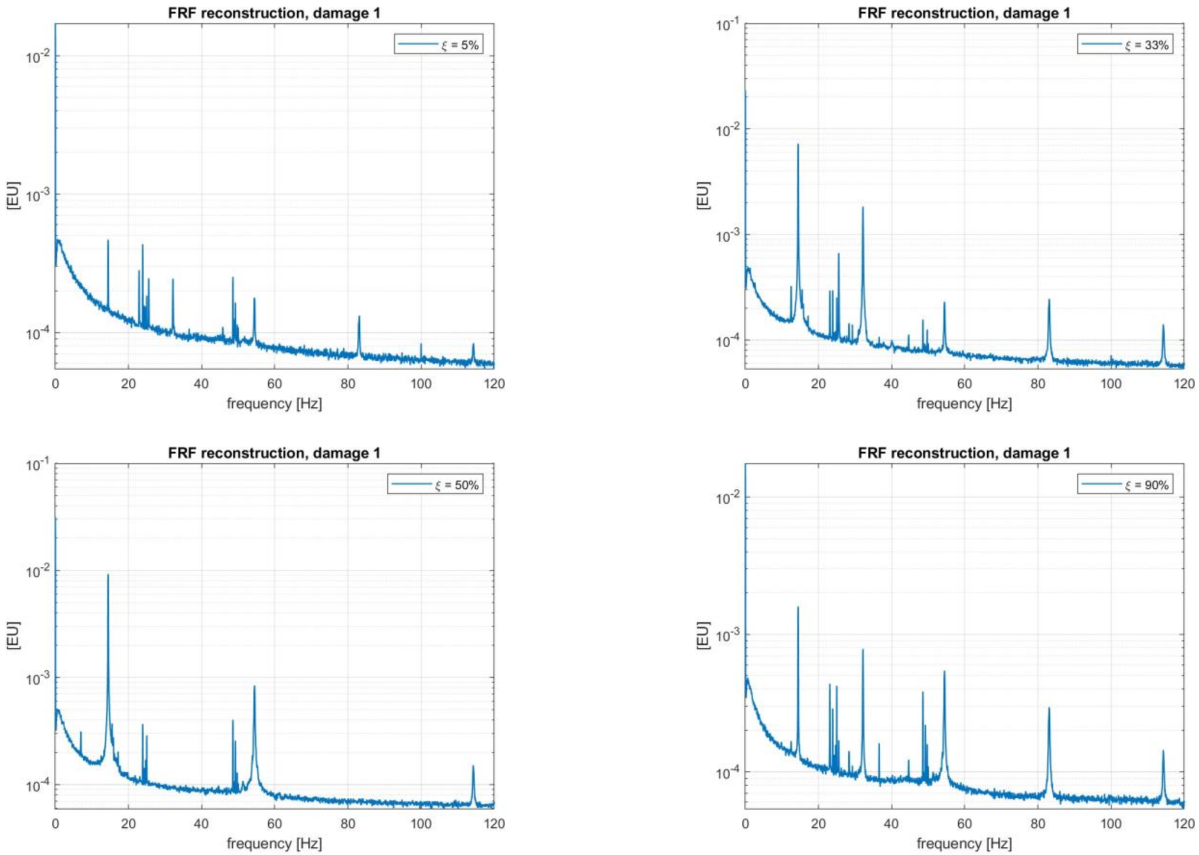



*Figure: reconstruction of the FRFs related to each sensor, damage 1*

# FRFs reconstruction

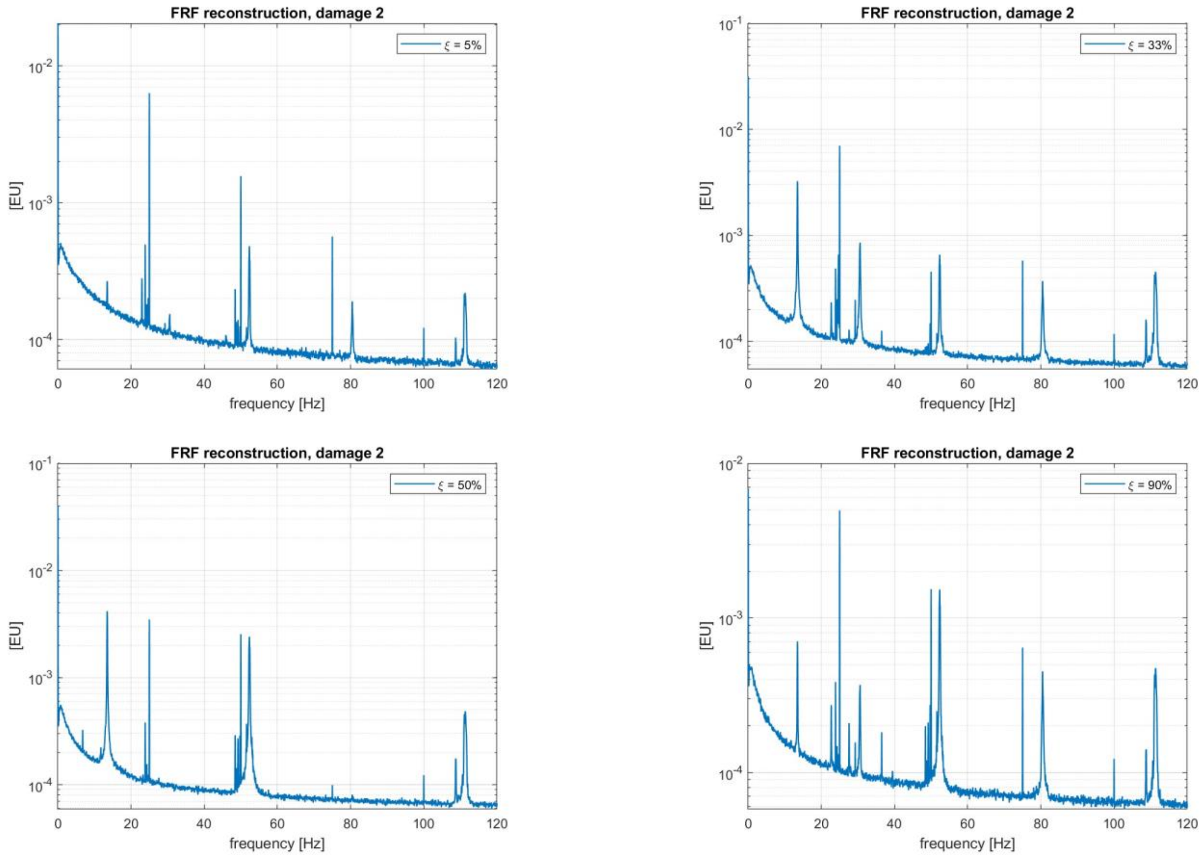


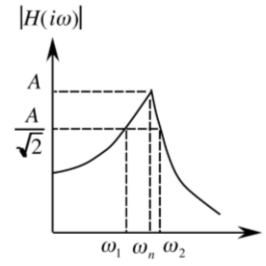

*Figure: reconstruction of the FRFs related to each sensor, damage 2*

# Damping estimation

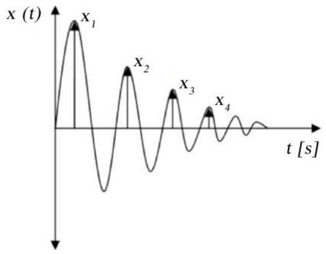
Once both natural frequencies and mode shapes have been properly estimated, modal damping identification procedure has been taken into account.

Two methods have been proposed for the $\xi_r$ identification:

- Half peak power method:




- Logarithmic decrement method



# Half Peak Power method

The half peak power method (or peak picking) is a technique commonly used for calculating the system damping using frequency response curves.

According with the OMA approach, since the first singular value curve well approximates the square of the FRF, we can apply the half power method directly on that curve.

Thus, considering the singular value curve about each eigenfrequency $\omega_r$, $\xi_r$ can be estimated as:

$$\xi_r = \frac{\omega_2^2 - \omega_1^2}{4\omega_r^2}$$

| | Healthy | Damage 1 | Damage 2 |
|---|---|---|---|
| $\xi_1$ | 0.17% | 0.12% | 0.40% |
| $\xi_2$ | 0.12% | 0.12% | 0.26% |
| $\xi_3$ | 0.095% | 0.20% | 0.13% |
| $\xi_4$ | 0.12% | 0.12% | 0.075% |

# Half Power method

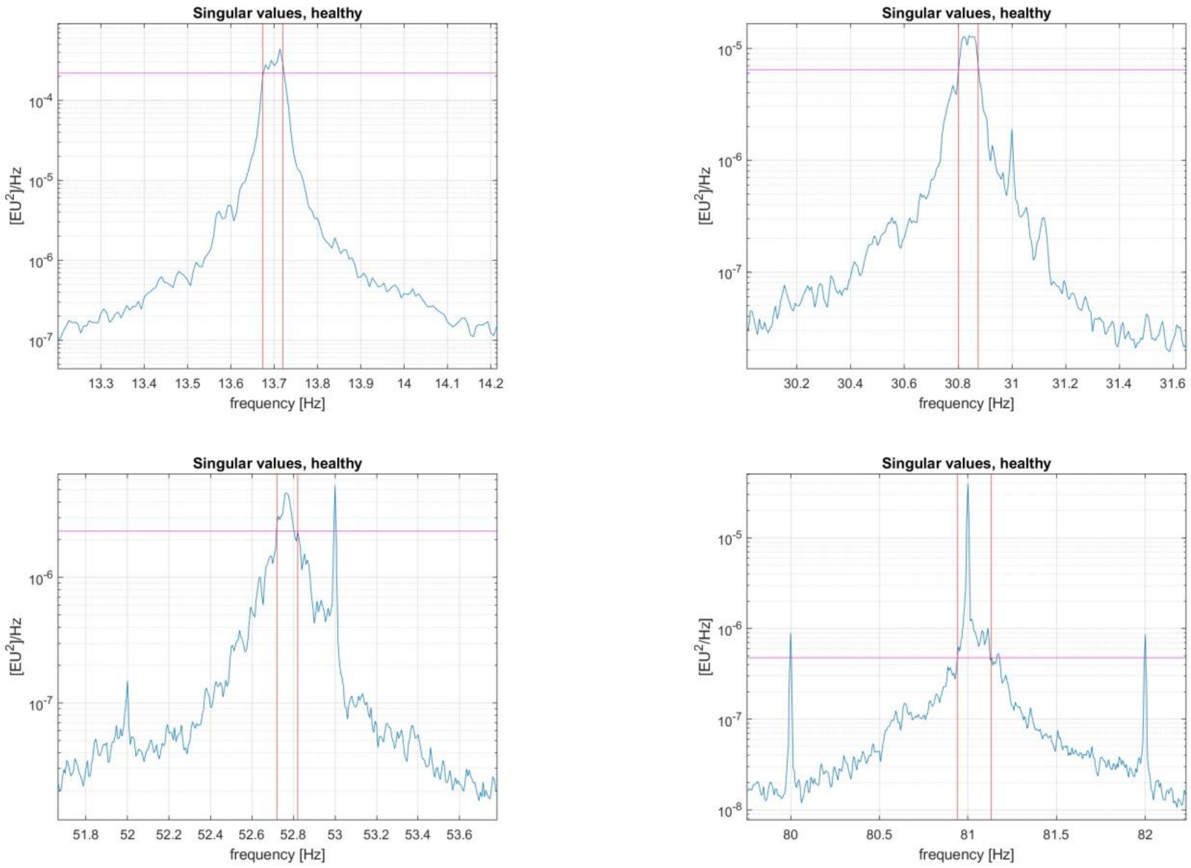



*Figure: half peak power method parameters in the resonance peaks, healthy condition*

# Logarithmic decrement method

Another method to estimate $\xi_r$ exploits the IRFs obtained by restricting the first singular value curves to a SDOF curve and applying the Inverse Fourier Transform.

By measuring the logarithmic decrement between two consecutive peaks it is possible to obtain a damping estimate:

$$\delta = \ln\left(\frac{h(t)}{h(t+T)}\right) = \frac{2\pi\xi_r}{\sqrt{1-\xi_r^2}} \quad \rightarrow \quad \xi_r = \frac{\delta}{\sqrt{4\pi^2+\delta^2}}$$

In order to have a more reliable result, the values obtained (for each peak) have been exploited as initial guesses for a least square error fitting with an exponential curve:

$$h(t) = Ae^{-\xi_r\omega_r t}$$

However, the results obtained were quite different from the expected ones.

That is due to the imprecise shape of the peaks of the singular value curves, indeed also the obtained IRFs proved to be non reliable.

# Logarithmic decrement method

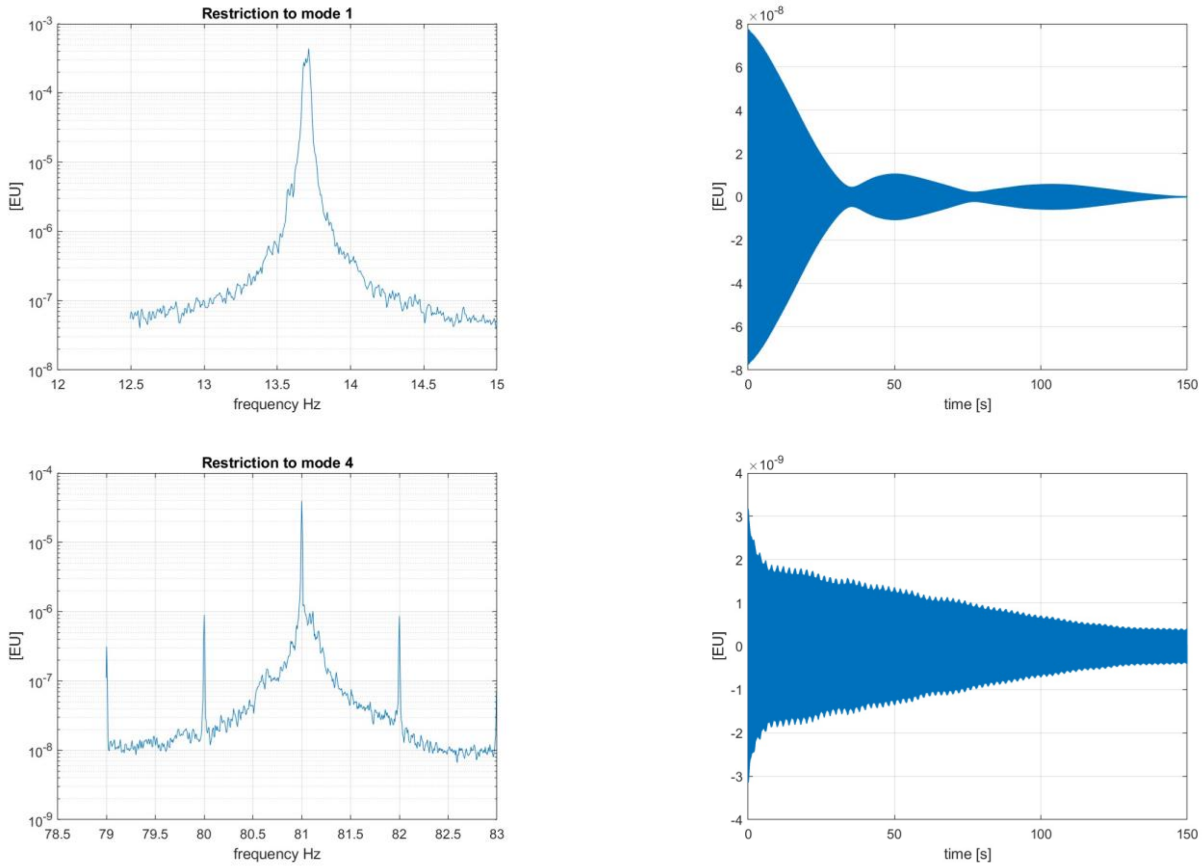



*Figure: resonance peaks and corresponding IRFs*

# Conclusions and final remarks

Once mode shapes and modal parameters are extracted, their variations can be observed for establishing whether the system is damaged or not.

The effect of damage can be distinguished from the effect of a change in the environmental conditions by considering together modal parameters of different modes.

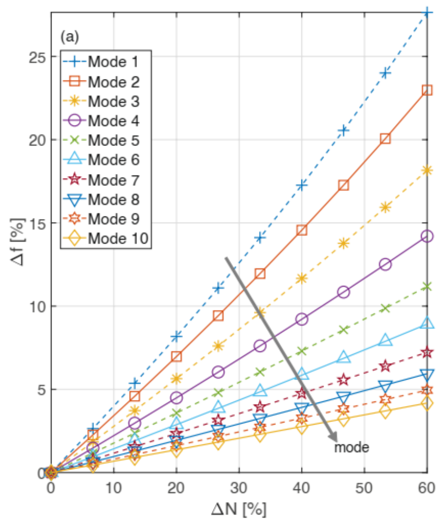



*Figure: eigenfrequencies trend as a function of axial load variation*

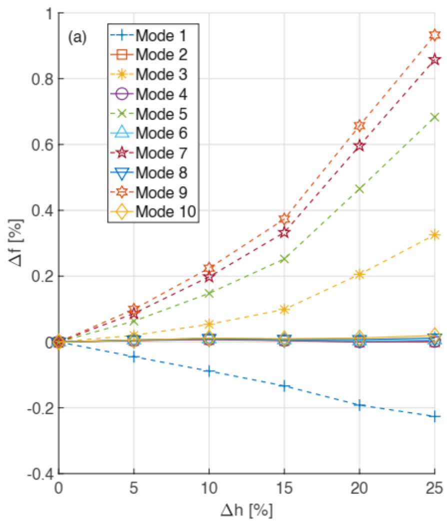



*Figure: eigenfrequencies trend as a function of damage variation*

# Conclusions and final remarks

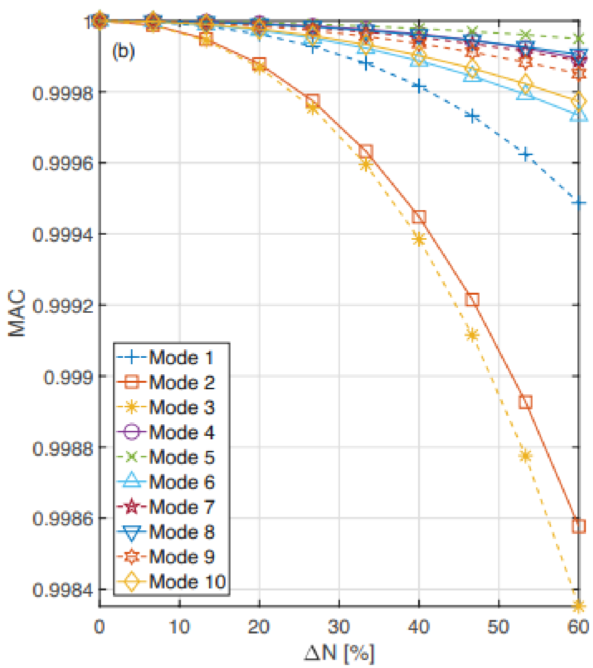



*Figure: MAC trend as a function of axial load variation*

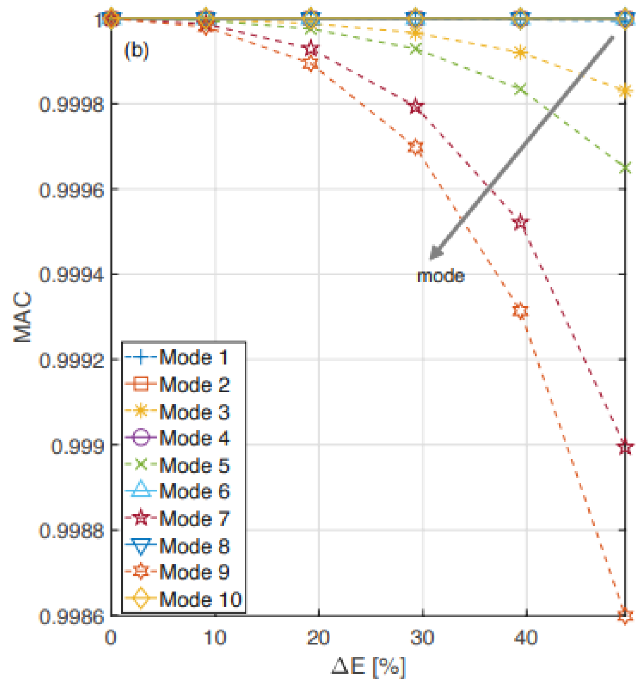



*Figure: MAC trend as a function of damage variation*

# Conclusions and final remarks

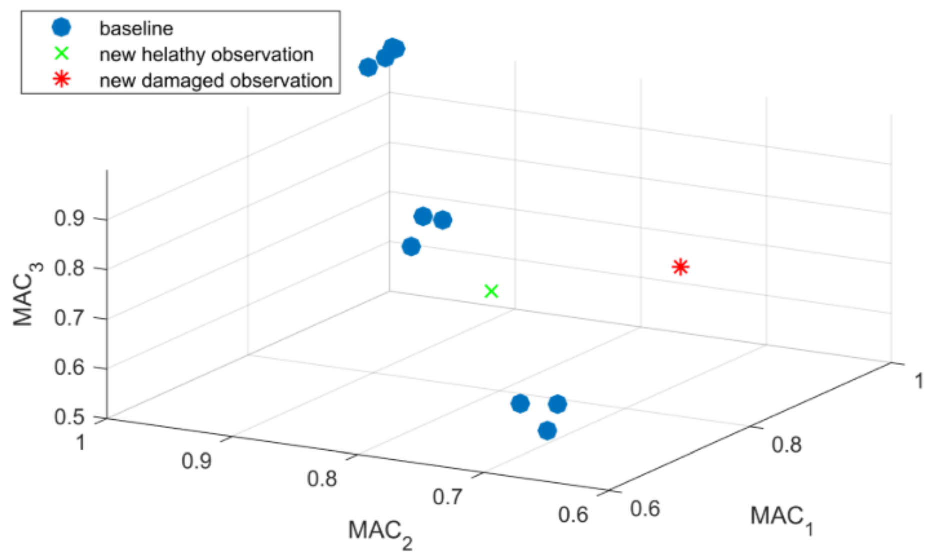



*Figure: baseline for building a machine learning algorithm able to identify damage using MAC values*

# Conclusions and final remarks

In the absence of sufficient data and of a properly constructed baseline for sana conditions, the trend of extracted modal parameters can be used just to verify if they're reliable or not.

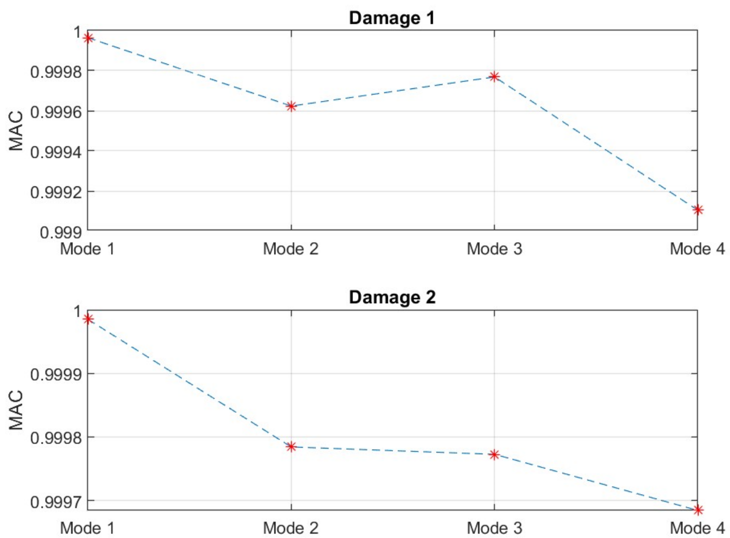



*Figure: MAC trend between different modes*

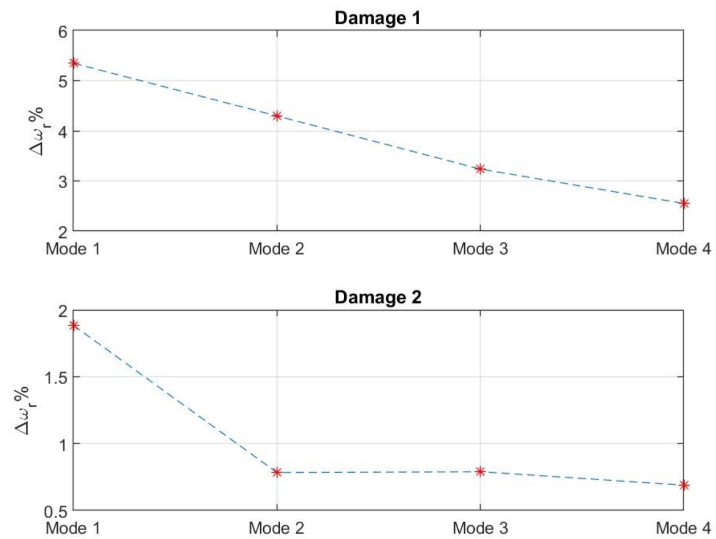



*Figure: eigenfrequencies shifting in percentage between different modes*

# Conclusions and final remarks

Key Advantages in monitoring a tie-rod using Operational Modal Analysis techniques:

- Continuous Health Monitoring
- Direct Extraction of Modal Parameters
- Enhanced Automation
- Early Fault Detection
- Cost-Effective and Efficient